

Frequency Response Measurement of Half-Wave Voltage and Chirp Parameter of LiNbO₃ Intensity Modulators in Low Frequency Range

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Abstract—We measured frequency responses of half-wave voltage V_π and chirp parameter α_0 of LiNbO₃ modulators below 1 GHz range using optical heterodyne scheme with an acousto-optic modulator. The coefficients of variation of the measured values of V_π and α_0 are 0.2 % and 2 %, respectively. We observed strong dependency of V_π and α_0 on the frequency.

Keywords—electrooptic modulators; intensity modulation; optical fiber communication; frequency response

I. INTRODUCTION

Measurement of half-wave voltage V_π and chirp parameter α_0 is important to characterize electro-optic modulators. In order to measure V_π and α_0 , optical spectrum of the modulator output is typically observed with an optical spectrum analyzer and the intensity of optical sidebands are measured [1-3]. However, it is difficult to measure the optical sidebands separately if modulation frequency is not quite high, e.g. below GHz region. Acousto-optic effect [4, 5] which usually appears in this frequency range in LiNbO₃ modulators is known as it influences system performance of analogue optical fiber link. Recently, frequency response measurement of V_π of a LiNbO₃ modulator in low frequency region was reported using optical heterodyne scheme with two independent wavelength tunable continuous wave (CW) lasers [6]. In this manuscript, we measure frequency response of V_π and α_0 of LiNbO₃ modulators using optical heterodyne scheme with a CW laser and an acousto-optic modulator (AOM) as an optical frequency shifter [7].

II. MEASUREMENT PRINCIPLE

Figure 1 (a) shows typical configuration for observing optical sidebands of the modulator output. A CW light, where the frequency is denoted by f_0 , is fed to a LiNbO₃ modulator (LNM). The LNM is driven by RF sine wave with the frequency of f_m and a dc bias voltage. The n -th order optical sidebands are appeared at $f_0 \pm n f_m$, where n is the integer number. The optical sidebands are measured with an optical spectrum analyzer (OSA). Therefore, if f_m is not very high, the OSA with very narrow resolution bandwidth is required for separately observing the optical sidebands. Figure 1 (b) shows the configuration used in our experiment, where the

configuration is based on a heterodyne interferometer. The CW laser output (f_0) is divided into two arms. The optical frequency of the light travels in one arm is shifted by f_A with an AOM. The light of the other arm is modulated with a LNM. The lights are combined and detected with a photodiode PD. The RF spectrum of the PD output is observed with an RF spectrum analyzer (RFSa). The optical sideband at $f_0 \pm n f_m$ is appeared at $|f_A \mp n f_m|$ at the RFSa. In this setup, the RF sideband power is proportional to the optical sideband power, since RF beat power is proportional to the product of the power of the two light. In typical heterodyne measurement, f_m is set to be much less than f_A . Then the whole sidebands are appeared around f_A as shown in Figure 2 (a). In this case, the maximum measurable f_m is f_A/n . Since f_A is typically around 100 MHz, frequency characteristics at more than several tens of MHz cannot be measured. When $f_m > f_A$, most of the sidebands are folded up to higher frequency region as shown

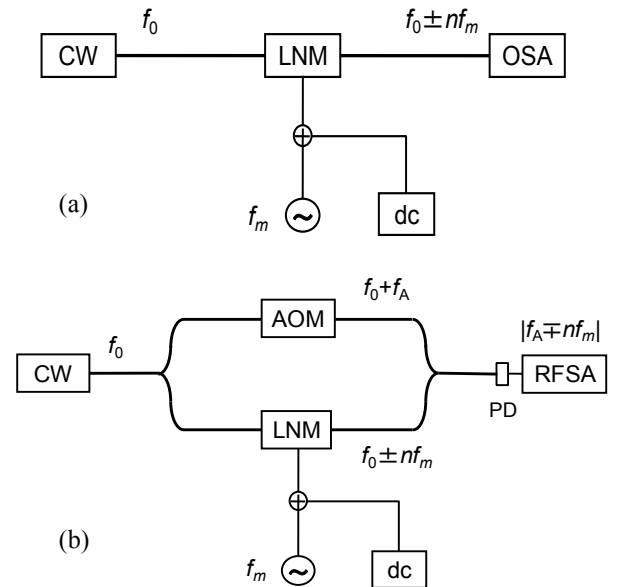


Figure 1. Experimental setup for observing optical sidebands of modulator output. (a) is a typical setup, where optical spectrum analyzer (OSA) is used. (b) shows the configuration used in the experiment.

in Figure 2 (b). However, if the sideband frequencies are not exactly equal, the sideband power can be measured. In this case, the flatness of the frequency response of the measuring equipment (PD and RFSA) is important. We checked the frequency response of PD and RFSA by measuring the power of RF beat with two independent wavelength tunable lasers. The flatness was ± 0.25 dB from 1 MHz to 3 GHz as shown in Figure 3. Therefore, the f_m of up to 1 GHz can be used with ± 0.25 dB accuracy if 3rd-order sideband is measured. The

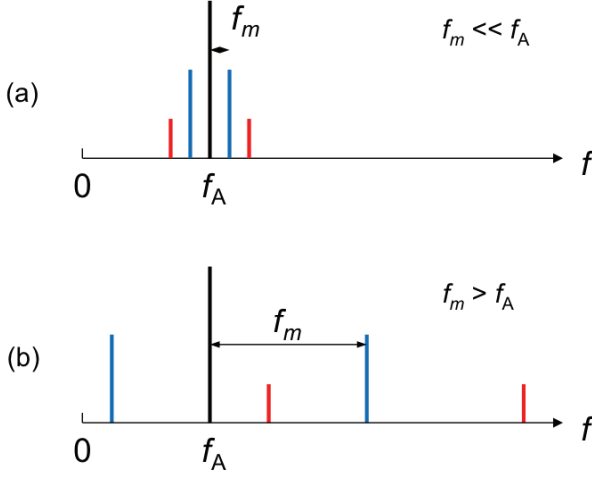


Figure 2. RF spectrum of the PD output, where (a) $f_m \ll f_A$, and (b) $f_m > f_A$.

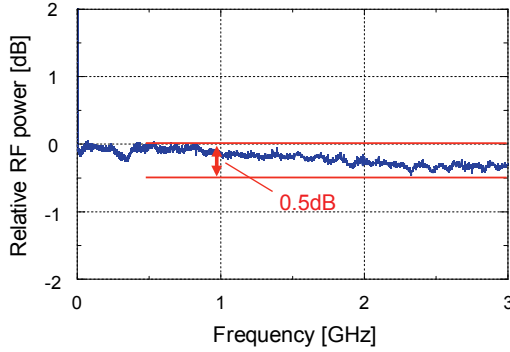


Figure 3. Flatness of the frequency response of PD and RFSA of the experimental setup using two individual wavelength tunable laser.

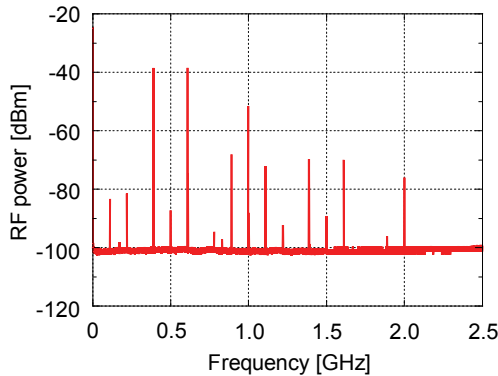


Figure 4. Observed RF spectrum with the RFSA when f_m and f_A is 500 MHz and 110 MHz, respectively.

observed RF spectrum, when f_m and f_A are 500 MHz and 110 MHz, respectively, is shown in Figure 4. We set the dc bias voltage of the modulator at null point. Each sideband can be observed separately.

III. RESULTS

In the experiment, we first measured the frequency response of the driving RF power. Figure 5 shows the result, where about 1.5 dB variations was observed. This variation is taken into account when V_π and α_0 are calculated. From this data and the frequency response of the input impedance of the modulator, the applied RF voltage to the modulator V_{RF} is calculated. Figure 6 shows the frequency response of the sideband power of the modulator output measured by the RFSA, when driving RF power is 26 dBm. In the measurement, we set the modulator to the null dc bias point, therefore even order sidebands are minimized. Now, if the measured 1-st, 2-nd and 3-rd order sideband powers are denoted by P_1 , P_2 and P_3 , the phase shift of the modulator's arm A obeys the following equations.

$$\sqrt{\frac{P_3}{P_1}} = \frac{J_3(A)}{J_1(A)}, \quad \sqrt{\frac{P_2}{P_1}} = \alpha_0 \left[A - \frac{2J_2(A)}{J_1(A)} \right] \dots (1)$$

A is solved from the first equation of Equation (1), and then α_0 is solved from the second equation of Equation (1). Once A is obtained, V_π can be calculated with Equation (2).

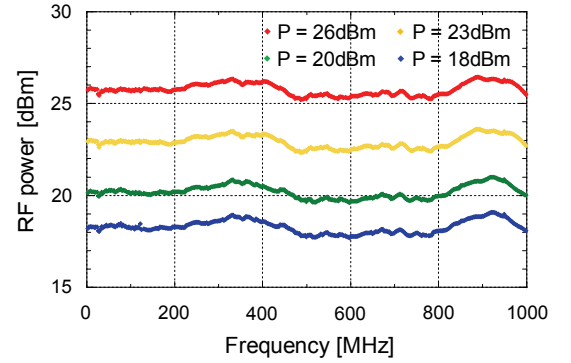


Figure 5. Frequency response of RF driving power to the modulator.

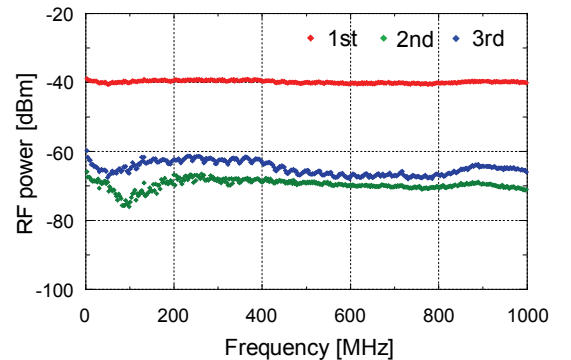
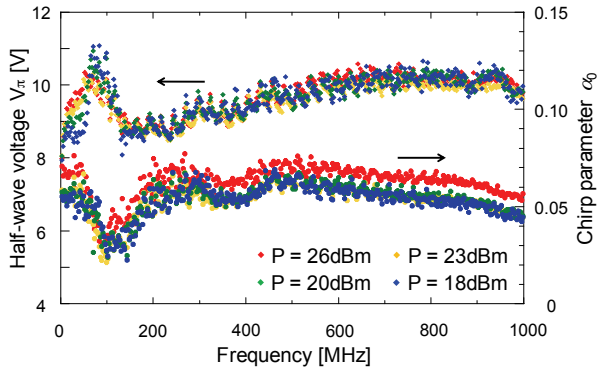
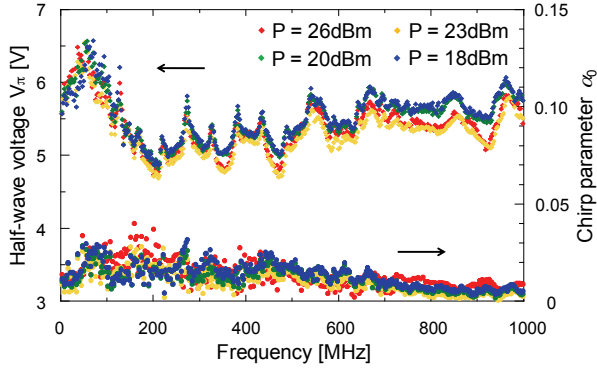


Figure 6. Frequency response of the observed sidebands of the modulator output with the RFSA.



(a) modulator A



(b) modulator B

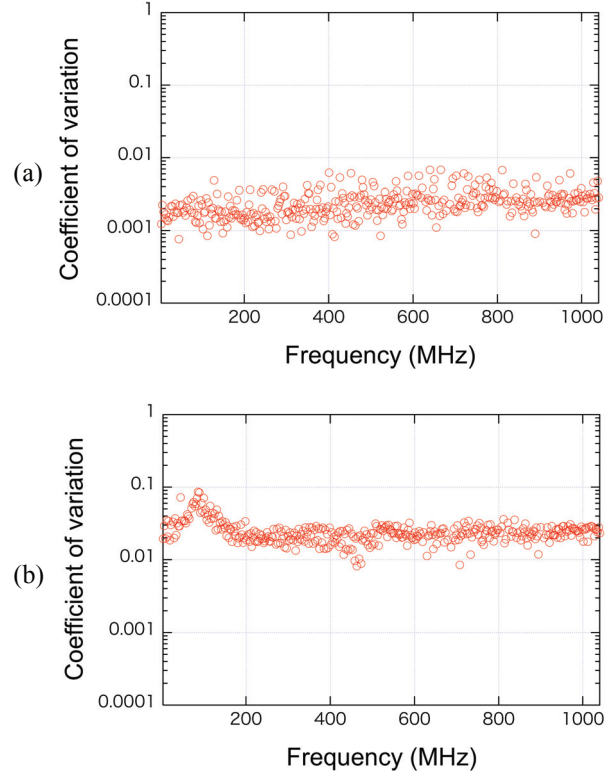
 Figure 7. Measured frequency response of LiNbO₃ modulators.

$$V_{\pi} = \frac{\pi V_{RF}}{2A} \dots\dots\dots (2)$$

In this way, we measured the frequency responses of V_{π} and α_0 up to 1 GHz with two LiNbO₃ modulators. Figures 7 (a) and (b) show the measured results. V_{π} and α_0 are almost independent of the driving RF power. However, strong dependencies of V_{π} and α_0 on the frequency are observed as shown in the figures. This may be mainly attributed to the acousto-optic effect of the substrate. In order to find out the accuracy of our measurement, we made 10 measurements of V_{π} and α_0 for modulator A with 26 dBm driving RF power, and obtained standard deviation of the measured values. Figures 8 show the coefficients of variation (CVs), which is the standard deviation normalized by the mean value, for V_{π} and α_0 . Averaged CVs over the entire frequency range are 0.2 % for V_{π} measurement and 2 % for α_0 measurement. CV of the α_0 measurement is larger since the 2-nd order sideband level is smaller therefore the measurement is influenced by the noise floor of the equipment.

IV. CONCLUSION

We measured the frequency response of half-wave voltage V_{π} and chirp parameter α_0 of LiNbO₃ modulators using optical heterodyne scheme with an acousto-optic modulator. We observed strong dependency of V_{π} and α_0 on the frequency. Coefficients of variation of the measured values are 0.2 % for V_{π} and 2 % for α_0 . We believe that accurate characterization


 Figure 8. Coefficients of variation of the measured values of V_{π} (a) and α_0 (b) for 10 measurements.

of the modulation parameters is important for developing electro-optic modulators for advanced multi-level modulation format such as QAM. Future work includes how these frequency dependencies influence on the quality of the modulator output.

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